SMART RESTAURANT MANAGEMENT SYSTEM

Software Requirements Specification

Functional and non-functional requirements

DOCUMENT ID	SRMS-05
VERSION	1.0
DATE	06 November 2026
STATUS	Draft for review
PHASE	Phase 1: event-driven core | Phase 2: AI camera detection
PURPOSE	Formal, numbered requirements with priorities, interfaces, use cases and acceptance criteria.

1. Introduction

1.1 Purpose

This Software Requirements Specification describes the functional and non-functional requirements of the Smart Restaurant Management System. It is intended for the project owner, designers, developers and testers.

1.2 Scope

The system allows guests to order and pay through a table QR code and gives restaurant staff and managers real-time control of orders, tables, alerts and cleaning. Phase 1 is event-driven. Phase 2 adds AI camera detection feeding the same table state engine.

1.3 Definitions

Term	Definition
SRS	Software Requirements Specification
QR session	A temporary guest session created by scanning a table QR code
State engine	Backend component owning table status transitions
KDS	Kitchen Display System
Must / Should / Could	Priority: Must is required for MVP, Should is important, Could is optional

1.4 References

- Project Overview (SRMS-01), User Roles (SRMS-02), Core Modules (SRMS-03), Database Schema (SRMS-04).
- Source inspiration: video on AI-assisted restaurant floor management (camera-based table status, wait and dwell tracking).

2. Overall Description

2.1 Product Perspective

A web-based system with four client interfaces (guest PWA, manager dashboard, staff apps for waiter/kitchen/cleaner, admin panel) sharing one backend, database and real-time channel. It integrates with a payment gateway and, in Phase 2, with camera streams.

2.2 User Classes

Customer, Waiter, Kitchen Staff, Cleaner, Manager, Owner/Admin as defined in SRMS-02.

2.3 Operating Environment

- Guests: current mobile browsers (Chrome, Safari, Samsung Internet) on Android 8+/iOS 14+.
- Staff: Chrome or Edge on tablet, phone or PC. Kitchen display runs full screen on a tablet or monitor.
- Server: Linux, containerised, managed PostgreSQL and Redis, HTTPS everywhere.

2.4 Constraints and Assumptions

- Internet connectivity is required in the restaurant; staff apps tolerate short drops with automatic resync.
- Payments must go through a PCI-DSS compliant gateway; the system never stores card data.
- Billing must comply with Indian GST invoice requirements.
- Phase 2 camera analysis must not store or recognise faces.

3. Functional Requirements

3.1 QR and Session (M1)

ID	Requirement	Priority
FR-QR-01	Each table shall have a unique, unguessable QR token that opens the menu for that table.	Must
FR-QR-02	Scanning shall create a session if none is active, or join the active session, and mark the table Occupied.	Must
FR-QR-03	The system shall allow only one active session per table.	Must
FR-QR-04	Sessions shall expire after payment, configurable idle time, or manager closure.	Must
FR-QR-05	Admin shall generate and print QR sheets and rotate a table token.	Should
FR-QR-06	Guests may enter guest count and optional phone number.	Could

3.2 Menu (M2)

ID	Requirement	Priority
FR-MN-01	The guest shall browse categories and items with image, price, description and veg marker.	Must
FR-MN-02	The system shall support variants and modifier groups with min/max rules.	Must
FR-MN-03	Sold-out items shall be unavailable to order immediately after staff mark them.	Must
FR-MN-04	The guest shall search and filter the menu.	Should
FR-MN-05	The menu shall support multiple languages.	Could

3.3 Cart and Ordering (M3)

ID	Requirement	Priority
FR-OR-01	Guests at one table shall share one cart synchronised in real time.	Must
FR-OR-02	The server shall compute all prices, taxes and totals; client values are not trusted.	Must
FR-OR-03	A guest shall be able to place multiple order rounds in one session.	Must
FR-OR-04	Placing an order shall create kitchen tickets and notify the assigned waiter.	Must
FR-OR-05	The guest shall see per-item live status.	Must
FR-OR-06	Items may be cancelled by the guest until the kitchen accepts; afterwards by staff only.	Should
FR-OR-07	Staff shall place or edit orders on behalf of guests.	Must

3.4 Kitchen (M4)

ID	Requirement	Priority
FR-KT-01	The KDS shall display new tickets in real time ordered by time with table, items, modifiers and notes.	Must
FR-KT-02	Kitchen staff shall set item status Accepted, Preparing, Ready.	Must
FR-KT-03	The KDS shall highlight tickets exceeding target preparation time.	Should
FR-KT-04	Kitchen staff shall mark items out of stock.	Must

3.5 Billing and Payment (M5)

ID	Requirement	Priority
FR-PY-01	The system shall maintain one running bill per session aggregating all orders.	Must
FR-PY-02	The bill shall show GST breakdown, optional service charge, discounts and tip.	Must
FR-PY-03	The guest shall pay online via UPI or card through the payment gateway.	Must
FR-PY-04	Payment confirmation shall be taken from verified gateway webhooks, not only from the client.	Must

ID	Requirement	Priority
FR-PY-05	Payment requests shall be idempotent to prevent double charge.	Must
FR-PY-06	Staff shall record cash payments.	Must
FR-PY-07	Discounts and refunds shall require manager approval and be audited.	Must
FR-PY-08	The system shall generate a sequential tax invoice and shareable receipt.	Must
FR-PY-09	The system shall support equal bill split and partial payment.	Should

3.6 Table Management and Floor Map (M6)

ID	Requirement	Priority
FR-TB-01	The system shall maintain table status: Available, Reserved, Occupied, Payment Pending, Needs Cleaning, Cleaning In Progress, Out of Service.	Must
FR-TB-02	Status transitions shall be performed only by the state engine according to the defined rules (SRMS-06).	Must
FR-TB-03	The manager dashboard shall show all tables on a floor map with colour, label, guest count, timer and alert icon, updating within 2 seconds.	Must
FR-TB-04	The dashboard shall show occupancy KPIs: tables occupied, guests seated, capacity percent, average dwell time, open alerts.	Must
FR-TB-05	Managers shall override a table state with a mandatory reason, logged.	Must
FR-TB-06	Every status change shall be written to the status log with actor and time.	Must
FR-TB-07	Owner shall design floors and table layout visually.	Should
FR-TB-08	Staff shall seat walk-in guests manually.	Must

3.7 Alerts (M7)

ID	Requirement	Priority
FR-AL-01	Guests shall be able to trigger Call Waiter and Request Bill alerts.	Must
FR-AL-02	The system shall automatically raise alerts for delayed orders, long wait without ordering, unattended table, long stay and payment failure.	Must
FR-AL-03	Alert thresholds shall be configurable per restaurant.	Must
FR-AL-04	Alerts shall be routed to the assigned waiter and escalate to the manager if not acknowledged within a set time.	Must
FR-AL-05	A table with any open alert shall display Needs Attention on the floor map.	Must
FR-AL-06	Alert acknowledge and resolve times shall be recorded.	Must

3.8 Cleaning (M8)

ID	Requirement	Priority
FR-CL-01	A cleaning task shall be created automatically when a session closes.	Must
FR-CL-02	Cleaners shall see a queue ordered by priority and age.	Must
FR-CL-03	Cleaners shall start and complete tasks; completion sets the table Available.	Must
FR-CL-04	Cleaners shall mark a table Out of Service with a note.	Should
FR-CL-05	The system shall record cleaning duration.	Should

3.9 Authentication, Staff and Admin (M9, M10)

ID	Requirement	Priority
FR-AU-01	Staff shall log in with PIN (floor roles) or email and password (manager, owner).	Must

ID	Requirement	Priority
FR-AU-02	Access shall be enforced by role as per the permission matrix in SRMS-02.	Must
FR-AU-03	Owner shall manage staff, roles, menu, tables, taxes, thresholds and gateway keys.	Must
FR-AU-04	Manager shall manage shifts and table assignment.	Should
FR-AU-05	Sensitive actions shall be audit logged.	Must
FR-AU-06	Menu shall be importable by CSV.	Could

3.10 Reports (M12) and Feedback/Waitlist (M13)

ID	Requirement	Priority
FR-RP-01	Reports shall cover sales by day/hour/category/item and by payment method.	Must
FR-RP-02	Reports shall cover table turnover, dwell time, cleaning time, alert response time and prep time.	Should
FR-RP-03	Reports shall export to CSV and PDF.	Should
FR-FB-01	Guests shall be able to rate their experience after payment; low ratings alert the manager.	Should
FR-WL-01	Staff shall manage a waiting list and notify guests by SMS or WhatsApp when a table is ready.	Could
FR-WL-02	Staff shall create reservations that mark tables Reserved.	Could

3.11 Phase 2: AI Camera Detection (M14)

ID	Requirement	Priority
FR-AI-01	The system shall map camera views to table zones.	Must (Phase 2)
FR-AI-02	The AI service shall detect people and report seated, left and cleared events with confidence scores.	Must (Phase 2)
FR-AI-03	Events above the configured confidence shall update the state engine; others shall appear as staff suggestions.	Must (Phase 2)
FR-AI-04	The system shall estimate entrance queue size and guest wait time.	Should (Phase 2)
FR-AI-05	The system shall flag tables with guests waiting for staff beyond a threshold.	Should (Phase 2)
FR-AI-06	The system shall not perform face recognition or store identifying images.	Must (Phase 2)
FR-AI-07	Manual overrides shall always take priority over AI events.	Must (Phase 2)

4. External Interface Requirements

Interface	Requirement
User interface	Mobile-first guest PWA; responsive dashboards; kitchen display optimised for distance reading; accessible colours plus icons (not colour alone)
Payment gateway	REST API and signed webhooks (Razorpay or equivalent); UPI, cards, wallets
Messaging	SMS/WhatsApp provider for receipts and waitlist; web push for staff
Printers (optional)	Kitchen or bill printer through browser print or network printer integration
Cameras (Phase 2)	RTSP streams processed by an edge or cloud AI service that posts signed events to the API

5. Non-Functional Requirements

Category	Requirement
Performance	Menu loads in under 2 seconds on 4G; floor map updates within 2 seconds of an event; API p95 under 300 ms for standard calls
Capacity	Support at least 100 tables and 500 concurrent guest sessions per outlet without degradation

Category	Requirement
Availability	99.5% monthly uptime target; graceful degradation: ordering continues, cash and manual flows remain available
Reliability	No lost orders; orders saved before acknowledgement; automatic reconnect and state resync for real-time clients
Security	HTTPS/TLS, hashed credentials, role checks on every request, rate limiting, input validation, protection against QR token guessing, encrypted secrets, OWASP Top 10 controls
Payments	No card data stored; webhook signature verification; idempotency; reconciliation report
Privacy	Compliance with India DPDP Act principles; minimal personal data; AI phase stores no faces
Usability	Guest can order within 3 taps from scan to cart; staff actions in one tap; large touch targets; English plus Hindi/Gujarati
Compatibility	Latest two versions of major mobile and desktop browsers
Maintainability	Modular codebase, automated tests, CI/CD, documented API, structured logging and monitoring
Scalability	Stateless API behind a load balancer; Redis for pub/sub; multi-outlet via restaurant_id
Auditability	Immutable table status log and audit log

6. Use Cases

UC-1 Guest orders and pays

- **Actor:** Customer. **Pre:** table has a QR code. **Post:** bill paid, table flagged for cleaning.
- **Main flow:** scan QR, see menu, add items, place order, track status, request bill, pay online, receive receipt.
- **Alternate:** guest chooses cash and the waiter records it. **Exception:** payment fails, so a Payment Failed alert is raised and the guest may retry or pay cash.

UC-2 Manager monitors the floor

- **Actor:** Manager. **Main flow:** open dashboard, see colours and timers, click a red-flagged table, view alert, assign or resolve, optionally override status.
- **Post:** alert resolved and logged.

UC-3 Cleaner resets a table

- **Actor:** Cleaner. **Pre:** table is Needs Cleaning. **Main flow:** open queue, start task, clean, tap Done.
- **Post:** table is Available and visible as green to the manager.

UC-4 Guest calls the waiter

- **Actor:** Customer. **Main flow:** tap Call Waiter, alert appears on the waiter phone and the table shows Needs Attention, waiter acknowledges and resolves.
- **Exception:** no acknowledgement within the threshold, so the alert escalates to the manager.

7. Acceptance Criteria (Phase 1 MVP)

- A guest can scan, order, and pay online without installing an app.
- The manager sees the table turn Occupied within 2 seconds of a scan.
- A Call Waiter tap shows Needs Attention on the map and on the waiter device within 2 seconds.
- Successful payment closes the session and moves the table to Needs Cleaning.
- Cleaner Done moves the table to Available and logs the duration.
- Prices on the bill match the menu with correct GST; invoice numbers are sequential.
- Duplicate payment attempts do not charge twice.
- All table state changes appear in the status log.

8. Traceability to Modules

Requirement group	Module	Document
FR-QR	M1	SRMS-03, SRMS-06
FR-MN	M2	SRMS-03, SRMS-08
FR-OR / FR-KT	M3, M4	SRMS-03, SRMS-06
FR-PY	M5	SRMS-04, SRMS-06
FR-TB / FR-AL / FR-CL	M6, M7, M8	SRMS-06, SRMS-09
FR-AU	M9, M10	SRMS-02, SRMS-07
FR-AI	M14	SRMS-10